

Tejas Yalamanchili

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EDUCATION

Purdue University

Aug 2024 - May 2026

Masters of Science, Computer Science (GPA: 3.77 / 4.00)

West Lafayette, IN

- **Achievements:** Dean's and Semesters List all Semesters
- **Coursework:** Compilers, Data Structures and Algorithms, Data Mining And Machine Learning, Intro to Artificial Intelligence, Operating Systems, Database Systems, Reinforcement Learning, Computer Vision Programming, Parallel Computing, Deep Learning

Purdue University

Aug 2021 - May 2025

Bachelors of Science, Computer Science (GPA: 3.87)

West Lafayette, IN

- **Achievements:** Deans and Semesters Honors List all semesters and 2 X hackathon winner
- **Coursework:** Operating Systems, Computer Networks, Databases, Systems Programming, Data Structures and Algorithms

SKILLS

- **Programming Languages:** Java, Go, Rust, Python, C/C++, C#, ARM Assembly, HTML, CSS, Javascript, SQL, CUDA, Typescript, GraphQL
- **Frameworks & Libraries:** Asp .Net, Spring Boot, Flask, React, Flutter, numpy, pandas, PyTorch, Tensorflow, Xcode
- **Tools & Technologies:** Azure, AWS, Hugging Face, GitHub, DagsHub, Linux, Kubernetes, Docker

EXPERIENCE

Cummins | Graduate Research Assistant

Aug 2025 - Present

- Lead a collaborative Purdue Data Mine-Cummins project to build predictive-maintenance AI pipelines using Python and Docker on real equipment data, from feature engineering and model training to validation on real failure cases.
- Designed an AI-driven agent system leveraging predictive-maintenance model outputs to detect failures, map them to affected components, and trigger automated supplier part-ordering workflows ahead of warranty expiration, improving reliability and reducing at-risk warranty expirations to ~20%.

Software Engineer Intern

May 2024 - Present

- Developed a robust engineer monitoring interface using Spring Boot and Docker, deploying on Kubernetes, integrating 5 new features like real-time tracking; reduced issue detection and resolution time by boosting system reliability and uptime.
- Deployed RAG and LlamaIndex-based chatbots with Hugging Face for technician assistance, improving troubleshooting processes through guided repair steps.

Tech Mahindra | Machine Learning Intern

May 2023 - Aug 2023

- Leveraged CODEBERT (LLM) and PyTorch as a key tool to consolidate and categorize similar functions within Nissan's automotive codebase into clusters, increasing overall developer productivity by 40% by developing a semantic code search to navigate through organized code clusters.
- Constructed a highly adaptable chatbot using RAG and large language models to enable effortless customization for enterprise search by empowering the chatbot to deliver context-aware responses, reducing search query resolution time by 25%.

Providence Global | Software Engineer Intern

Jun 2022 - Aug 2022

- Engineered a web API using .Net Core to facilitate seamless migration of keys and settings to Azure cloud and Azure App configuration achieving a 30% reduction in deployment time through this strategic migration.
- Optimized SQL database for client records by implementing techniques such as data normalization and indexing, reducing query time from 30 sec to 4 sec and improving overall database performance.

AUUG | Software Engineer Intern

May 2022 - Jun 2022

- Developed a C# library with .NET that streamed musical information to multiple devices, expanding cross-platform availability
- Harnessed the O2 communication protocol, ensuring efficient data transmission, ultimately enhancing user experience
- Optimized the application's performance, resulting in a remarkable 15% reduction in latency and significantly improving the overall responsiveness and reliability of the streaming service.

PROJECTS

AgentTrail | <https://agent-relay-website.vercel.app/>

Jan 2026 - Present

- Built and deployed a distributed, transaction-aware agent runtime library with structured run logging and API key management, enabling traceable failure recovery and observability with cost-management controls
- Developed and deployed a containerized production web platform using React, FastAPI, AWS (S3, CloudFront, Elastic Beanstalk) and Kubernetes, providing real-time dashboards, OAuth-secured access, and scalable agent monitoring

Fit Check

Jan 2024 - Mar 2024

- Pioneered an AI-powered e-commerce platform aggregating apparel from over 10 leading brands; integrated computer vision models in TensorFlow for virtual try-on, improving cloth mapping accuracy on sample avatars by 40%
- Managed full-cycle model deployment and web development using Hugging Face and Kubernetes, optimizing ML infrastructure and reducing page load time for enhanced user experience

UniLeap

May 2023 - Dec 2023

- Co-founded UniLeap to provide a personalized mentoring platform for high schoolers and graduate students needing college application and visa assistance for studying abroad
- Steered the deployment of Django backend on AWS EC2, and used Nginx as a load balancer and React.js frontend on AWS Amplify to provide optimal performance and a user-friendly experience for 10+ active students.